
DAWN-SRW: State-Conditioned Programs for Learned State Transitions

Seungho Choi¹

Abstract

Learned state-transition computation is often organizationally implicit. We ask whether it can be organized so that the conditions, operations, and transition effects constituting its execution are explicit objects of measurement and intervention. We study this question constructively in autoregressive language models. We use $(\mathcal{O}, C, \Gamma)$ as minimal organizational coordinates for one family of explicit learned transition organizations. Within these coordinates, a *State-Conditioned Program* is one program-oriented formulation in which controlled operation contributions are constituents of actual forward execution. It is not a claim of symbolic program text, formal operational semantics, or an intrinsic or privileged neural decomposition. DAWN-SRW provides one constructive realization using rank-1 read-write (RW) operators that read a scalar from the current representation and write along a learned transition direction, together with an operation-address space, normalized additive composition, full candidate scope, and dense RW execution. At 393.8M parameters and 39.999B tokens, DAWN-SRW supports viability relative to the matched dense control: C4 cross-entropy is 2.9402 versus 2.9359 and the six-task zero-shot mean is 0.4629 versus 0.4597. State-conditioned exact applicability support provides evidence of conditional organization. A discovery-frozen 4,096-site Q/K-centered operator circuit passes suppression, four matched-control families, and exact restoration on validation and held-out indirect-object-identification (IOI) inputs, establishing causal accessibility; fail-closed results delimit unsuccessful localization. They do not establish human-readable operator semantics, benchmark superiority, hardware efficiency, or an intrinsic or unique decomposition.

1. Introduction

Learned systems compute through state transitions, but their transition-producing organization is often implicit. State dependence and conditional structure are established; the broader problem is whether learned computation can be architecturally organized around explicit conditions, operations, and transition effects. Autoregressive language models provide a concrete test setting in which high-dimensional representations undergo repeated transitions. If X_ℓ denotes the sequence representation at layer ℓ , a block implements, at a coarse level, $X_\ell \mapsto X_{\ell+1}$.

We therefore ask:

Can learned state-transition computation be organized so that the conditions, operations, and transition effects constituting its execution are explicit objects of measurement and intervention?

Autoregressive language modeling is the present constructive test domain, not a boundary on this research question.

At the broadest level, program execution and dynamical systems share the substrate $x \rightarrow x'$: the former treats computation as state or configuration change, while the latter treats the same abstract structure as state evolution. This commonality does not give DAWN-SRW a formal operational semantics. A more specific organizational motif, condition $\rightarrow$ operation $\rightarrow$ state transition, is likewise old; we do not claim it as novel. To separate its descriptive roles, we use the organizational coordinates

$$\mathcal{P} = (\mathcal{O}, C, \Gamma), \quad (1)$$

where $\mathcal{O} = \{O_i\}_i$ is an operation family, C is the state-conditioned applicability relation, and Γ is the composition rule. For scalar applicability,

$$C(x, O_i) = c_i(x), \quad F_{\mathcal{P}}(x) = \Gamma(\{c_i(x)O_i(x)\}_i). \quad (2)$$

These minimal coordinates describe one family of explicit learned transition organizations, not a universal ontology or necessary decomposition of learned dynamics. The form does not itself state the architectural hypothesis. Program and execution traditions motivate identifying conditioned operations; dynamics and control provide the richer mathematical precedent for continuously valued validity and

¹Independent Researcher. Correspondence to: Seungho Choi <madst0614@gmail.com>.

composition of overlapping component effects. The coordinates distinguish what transformations exist, when and how much they apply, and how their controlled contributions form a transition-producing result. Within them, the *control-operation factorization* separates C from $\mathcal{O}$; Γ is a separate composition role, not a third factor of that factorization. Normalized additive composition is one DAWN-SRW realization of Γ , not part of the coordinates’ definition.

The coordinates alone are not the scientific claim. An arbitrary mapping $T(x)$ is vacuously compatible by taking one operation $O_1 = T$ and $c_1(x) = 1$. Such post-hoc functional decomposability is weaker than a learned system whose forward execution is actually organized through operation identities, state-conditioned applicability, and controlled contributions; a further architectural step exposes those constituents as explicit variables or objects. Our interest is in this computational organization and architectural exposure, not in algebraic decomposability by itself.

Within this broader problem, we call one program-oriented organizational formulation a *State-Conditioned Program*: the current state determines operation applicability, and controlled operation contributions are constituents of the actual transition-producing computation. This definition requires neither continuous or geometric control, sparse support, rank-1 operations, persistence, nor additive composition. The term *program* generalizes the analogy beyond symbolic source, discrete instructions, named human-readable primitives, call/return control flow, or formal operational semantics.

The same learned computation admits complementary descriptions. A neural-network view asks what tensor map $x \mapsto F_P(x) \mapsto x'$ is implemented; a dynamics view asks what evolution $x \mapsto x'$, or residual update $x \mapsto x + \Delta x$, occurs. The State-Conditioned Program view asks which operations contribute at a state, under what conditions, by how much, and what happens when selected contributions are changed. It is therefore a computational view grounded in actual execution, not an ontological claim that this is the only true, intrinsic, necessary, or privileged neural decomposition. Operational accounts of transitions between configurations provide an operation-oriented conceptual lineage (Plotkin, 1981), but no formal semantics for DAWN-SRW is claimed.

The resulting hierarchy is conceptual, not chronological. Its convergence point is explicitly organized learned state-transition computation, not the State-Conditioned Program formulation itself. Dynamics and control provide strong mathematical precedent for graded state-dependent validity, local transition models, and weighted composition (Zadeh, 1965; Takagi & Sugeno, 1985; Johansen & Foss, 1995; Rugh & Shamma, 2000). Learned conditional and modular systems then make both C_θ and $O_{\theta,i}$ learned computational

objects (Shazeer et al., 2017; Rosenbaum et al., 2018; Goyal et al., 2021b; Rahaman et al., 2021); stable operation identity is an architectural property, whereas reuse is an empirical possibility. Read-write and fast-weight work supplies an elementary transformation structure (Schlag et al., 2021; Lu et al., 2026). The architectural hypothesis studied here is that learned transition-producing computation can organize its actual forward execution through stable learned operations under explicit state-conditioned control over their contributions. We test that proposition constructively in autoregressive language modeling, not the novelty of any constituent ingredient.

Architectural exposure gives operation identities, applicability coefficients, operation responses, and controlled contributions direct measurement and intervention targets. In the tested protocol, selected contributions are causally accessible through suppression and exact restoration. This does not establish semantic purity, universal cross-context meaning, arbitrary operation replacement, control transplant, or program synthesis.

DAWN-SRW is one constructive realization. It uses persistent rank-1 RW operators, continuous-valued applicability in a $d_c = 256$ operation-address space, and normalized additive composition across Q/K/V and the residual-state transition (RST). Applicability is controlled through this low-dimensional representation, whereas each applicable RW operation still reads the original D -dimensional representation and writes in $\mathbb{R}^D$. Thus low-dimensional applicability control is not a low-dimensional content read. The evaluated implementation considers the full type-specific operator pool and executes RW contractions densely. Its spherical operation-address geometry is a control inductive bias. Neither geometry nor rank-1 structure, persistence, global candidate visibility, or additive composition is required by the formulation.

The empirical study tests this realization through three ordered claims: viability, conditional organization, and causal accessibility. Fail-closed results delimit where the mechanistic localization procedure does not succeed; efficiency is not a headline claim.

Our contributions are:

1. **An organizational formulation for explicit learned state-transition computation and a constructive architectural test.** We connect operation-oriented program execution, continuous state-dependent dynamics, and learned modular computation as complementary precedents; use $(\mathcal{O}, C, \Gamma)$ as minimal organizational coordinates; formulate the State-Conditioned Program as one operation-oriented organization of these roles in actual forward execution; and test the resulting architectural hypothesis through DAWN-SRW in autore-

gressive language modeling. DAWN-SRW provides one concrete realization using persistent rank-1 RW operations and low-dimensional continuous geometric applicability.

2. **A constructive empirical test with bounded evidence.** We test viability at 400M scale, report descriptive results for the 1.3B DAWN-SRW checkpoint (1.272B parameters), provide evidence of conditional organization through state-conditioned exact applicability support, and test causal accessibility through a discovery-frozen IOI circuit confirmed by suppression, matched controls, and exact restoration. Fail-closed results bound where the same localization procedure did not succeed.

2. Related Work

2.1. Operational views of computation and state transition

Structural operational semantics describes computation through transitions between configurations governed by rules (Plotkin, 1981). We use only the associated operation-oriented view: execution can be organized around identifiable operations and conditions under which they apply. DAWN-SRW does not define programming-language syntax, guarded commands, inference rules, loops, or formal operational semantics. Its overlap with dynamics begins at the abstract substrate $x \mapsto x'$, not at a claimed formal equivalence.

2.2. Continuous state-conditioned dynamics and local-model composition

Dynamical modeling supplies the primary mathematical lineage for conditional transition organization. Fuzzy sets formalized continuously graded membership rather than Boolean set inclusion (Zadeh, 1965). Takagi–Sugeno models couple graded premise validity to local consequent models and aggregate their effects (Takagi & Sugeno, 1985). Local model networks likewise interpolate simple component models, each with a limited operating regime, into a global nonlinear model (Johansen & Foss, 1995). Gain scheduling and linear parameter-varying formulations organize families of local laws or controllers by measured scheduling variables, while also making clear that properties of local designs do not automatically transfer to the varying global system (Rugh & Shamma, 2000).

Learned-dynamics research also makes transition structure an explicit learned object. Decomposed Linear Dynamical Systems learn dynamics components whose sparse combinations represent overlapping nonstationary dynamics (Mudrik et al., 2024); Neural Hybrid Automata learn discrete modes and events with multi-mode continuous dynamics

(Poli et al., 2021). Neither is a State-Conditioned Program or equivalent to DAWN-SRW; both provide precedent for explicit learned transition structure.

Together these traditions establish precedent for continuous state- or regime-dependent validity, overlapping component transition laws, and weighted effect composition. They are not mathematically identical to one another or to DAWN-SRW, and they need not expose persistent learned operations or causal intervention interfaces. State-conditioned dynamics, continuous membership, and weighted combinations of local models are therefore not novelty claims here.

2.3. Learned conditional and modular computation

Mixture-of-experts (MoE) computation already has the generic form

$$F(x) = \sum_i G_i(x) E_i(x), \quad (3)$$

with input-dependent control over learned transformations. Sparsely-gated MoE (Shazeer et al., 2017), Switch Transformer (Fedus et al., 2022), ModuleFormer (Shen et al., 2023), and CondConv (Yang et al., 2019) establish conditional routing and weighted expert combination. PEER retrieves million-scale tiny experts (He, 2024), while Omni-MoE uses vector-level Atomic Experts and structured routing (Shi et al., 2026). Continuous or differentiable control over learned transformations is therefore precedent; MoE-style conditional computation and continuous applicability are not claimed as novel.

Shared parameter identity across depth is likewise established: Universal Transformers recur over shared layers (Dehghani et al., 2019), MoEUT shares expertized feed-forward and attention computation (Csordás et al., 2024a), MoUE uses a layer-agnostic expert pool (Chen et al., 2026), and UniPool exposes a global pool through per-layer routers (Huang et al., 2026). SwitchHead routes attention-side experts (Csordás et al., 2024b), UMoE shares experts across attention and FFN computation (Yang et al., 2025), and Compositional Attention separates query–key search from value retrieval (Mittal et al., 2022). These architectures make reuse possible and measurable; stable or shared identity does not by itself establish broad functional reuse.

Routing Networks combine a router with learned function blocks and recursively choose functions from the current activation, producing input-specific dynamic function composition (Rosenbaum et al., 2018). Recursive Routing Networks apply the same organizing principle to language understanding through a shared set of learned functions and a meta-learner router (Cases et al., 2019). These are direct precedents for state/input-conditioned selection and composition of learned functions. Recurrent Independent Mechanisms use groups with independent transition dynam-

ics that compete to update when relevant and communicate through attention (Goyal et al., 2021b). The declarative/procedural factorization associated with the SCOFF line separates object-state files from reusable schemata that prescribe updates (Goyal et al., 2021a). Neural Production Systems learn condition–action rules whose state- and entity-dependent matches update entity state (Didolkar et al., 2021), and Reusable Slotwise Mechanisms dynamically select reusable mechanisms to predict future object-slot states (Nguyen et al., 2023).

These works expose learned mechanism identity and conditional use: RIM, SCOFF, Neural Production Systems, and Reusable Slotwise Mechanisms emphasize world or object-state dynamics, while Routing Networks and related executors emphasize neural-computation functions. This is emphasis, not a strict taxonomy or a claim that they solve the same problem.

Neural Interpreters use learned types and signatures to control access to reusable functions in a hyperspherical type space (Rahaman et al., 2021), and Neural Attentive Circuits separate a circuit generator from a modular executor (Weiss et al., 2022). These works establish conditional function access and dynamic learned configurations. In this layer, both C_θ and $O_{\theta,i}$ are learned objects: a local or component transition model can be interpreted architecturally as a persistent learned computational operation. Persistence supplies identity; whether that identity supports cross-context reuse is an empirical question, not part of the definition.

2.4. Read–write and low-rank operation structure

Geva et al. interpret Transformer feed-forward layers as key-value memories with input-sensitive coefficients over output directions (Geva et al., 2021), and later decompose their additive updates into parameter-vector sub-updates traced through vocabulary-space predictions (Geva et al., 2022). Product Key Memory provides lineage for factorized keys and structured retrieval (Lample et al., 2019), scalable memory layers extend content-addressable trainable memories (Berges et al., 2024), Fast Weight Programmers provide an outer-product programming precedent (Schlag et al., 2021), and MoRAM uses rank-1 associative-memory experts with intrinsic keys (Lu et al., 2026). Thus memory, retrieval, outer-product structure, read/write decomposition, and rank-1 experts are precedents. They provide the concrete transformation structure, not the general State-Conditioned Program definition. DAWN-SRW chooses $O_i(x) = \bar{w}_i(\bar{r}_i^\top x)$, then makes $c_i(x)O_i(x)$ an explicit controlled transition contribution under a separate applicability profile.

Implementation-level overlap in rank-1 experts, adaptive thresholds, geometric or low-dimensional control, read/write structure, or causal intervention does not imply shared conceptual ancestry, interpretation, or research

program.

2.5. Neural programming as adjacent precedent

Neural Programmer-Interpreters execute programs from supervised traces (Reed & de Freitas, 2016), Neural Programmer differentially selects operations and data (Neelakantan et al., 2016), and Fast Weight Programmers connect outer-product computation to instructions (Schlag et al., 2021). RASP gives a programming-language-level model of Transformer computation and realizability (Weiss et al., 2021). Neural programs, differentiable operation selection, and program-level descriptions of Transformer computation therefore have precedent. The State-Conditioned Program analogy is deliberately weaker and different: it concerns the organization of conditional learned state-transition operations, with no requirement for symbolic source, discrete instruction traces, human-readable predicates, named primitive semantics, or formal structural operational semantics.

2.6. Causal intervention as evaluation methodology

Mechanistic interpretability localizes components through causal intervention. IOI work isolates an attention-head circuit (Wang et al., 2023), AtP* approximates activation patching at scale (Kramár et al., 2024), and RAVEL evaluates localization against multiple causal criteria (Huang et al., 2024). This literature distinguishes candidate localization from causal confirmation. DAWN-SRW supplies controlled RW-operation contributions as intervention targets; its discovery freeze, suppression, matched controls, and exact restoration test the specified IOI circuit without implying general semantic interpretability. This work supplies a validation methodology, not a constructive ingredient of the architecture.

The traditions above are neither historically linear nor mathematically identical. Program and execution accounts contribute an operation-oriented view of conditioned computation. Dynamics and control provide the mathematical precedent for continuously state-dependent validity and composition of transition effects. Learned modular systems show that control and computational components can themselves be learned and assigned persistent identities. RW/fast-weight work supplies elementary transformation structures, while causal-intervention work provides methodology for testing whether exposed constituents matter to behavior. We connect these precedents at the level of learned state-transition computation itself: whether execution can be organized so that conditions, operations, and transition effects become explicit computational objects.

At the coordinate level, one family of such organizations can be expressed as

$$F_\theta(x) = \Gamma(\{c_{\theta,i}(x)O_{\theta,i}(x)\}_i),$$

using $(\mathcal{O}, C, \Gamma)$ as minimal organizational coordinates. This is an organizational form, not the architectural hypothesis and not a universal theory of learned dynamics. A State-Conditioned Program is one program-oriented formulation in which these roles participate in actual forward execution. The architectural hypothesis is the proposition that learned transition-producing computation can realize that organization through stable learned operations under explicit state-conditioned control over their contributions. DAWN-SRW is one concrete realization, at the granularity of a controlled RW contribution,

$$c_i(x)O_i(x) = c_i(x) \bar{w}_i(\bar{r}_i^\top x).$$

Applicability determines when and how much the operation contributes, the read direction extracts a state-dependent scalar from the current representation, and the write direction determines the resulting transition effect. The present constructive test domain is autoregressive language modeling; the broader problem is not limited to that domain.

3. DAWN-SRW

3.1. State Transitions and Control–Operation Factorization

Let

$$X_\ell = (x_{\ell,1}, \dots, x_{\ell,T}) \in \mathbb{R}^{T \times D}$$

denote the sequence representation at depth ℓ . A Transformer-like block produces state transitions through attention and a subsequent residual transformation:

$$Y_\ell = X_\ell + \text{Attn}_\ell(X_\ell), \quad (4)$$

$$X_{\ell+1} = Y_\ell + R_\ell(Y_\ell). \quad (5)$$

Equations 1–2 provide a coordinate-level organizational form, not a sufficient condition for computational organization or the architectural hypothesis itself. That hypothesis is that learned transition-producing computation can organize its actual forward execution through stable learned operations under explicit state-conditioned control over their contributions. DAWN-SRW exposes one realization through persistent RW operators, an explicit applicability profile, and normalized additive composition. For a token state $x \in \mathbb{R}^D$ and computation type $a \in \{Q, K, V, \text{RST}\}$, define

$$F_a : \mathbb{R}^D \rightarrow \mathbb{R}^D. \quad (6)$$

Here F_Q, F_K, F_V parameterize relational interaction and F_{RST} parameterizes a direct residual-state update. Rank-1 RW operators, operation-address geometry, persistent identities, full candidate scope, dense physical execution, and normalized additive composition are realization choices, not requirements of either the formulation or the architectural hypothesis. We use *operation* for the abstract transformation, *operator* for an indexed learned architectural object

implementing an operation, and *RW operator* for the rank-1 realization used here. Thus the computational formulation, architectural hypothesis, DAWN realization, and geometry implementation are distinct levels of claim.

3.2. Read-Write Operators

Each RW operator owns learned read and write parameters. With $\text{unit}(z) = z / \max(\|z\|_2, \epsilon)$, define normalized directions $\bar{r}_{a,i} = \text{unit}(r_{a,i})$ and $\bar{w}_{a,i} = \text{unit}(w_{a,i})$. The RW operator executes

$$O_{a,i}(x) = \langle x, \bar{r}_{a,i} \rangle \bar{w}_{a,i} = \bar{w}_{a,i} \bar{r}_{a,i}^\top x. \quad (7)$$

One controlled RW contribution can therefore be read as

$$c_{a,i}(x)O_{a,i}(x) = \underbrace{c_{a,i}(x)}_{\text{applicability}} \underbrace{(\bar{r}_{a,i}^\top x)}_{\text{scalar state read}} \times \underbrace{\bar{w}_{a,i}}_{\text{write direction}}. \quad (8)$$

Applicability and the read response are both state-dependent but serve different roles. Applicability determines whether and to what degree the operation contributes; the read direction extracts an operation-specific scalar operand from the current representation; and the write direction determines how that operand changes the state. Q/K/V RW operators compose attention interfaces; RST RW operators compose a direct residual-state update.

3.3. State-Conditioned Applicability

For each computation type, the current state induces an applicability profile over its N_a persistent RW operators,

$$C_a(x) = (c_{a,1}(x), \dots, c_{a,N_a}(x)), \quad (9)$$

whose coordinates determine whether and to what degree each RW operator participates. Section 3.5 defines the state-derived query $q_a(x)$ and operator address $k_{a,i}$. Their applicability score and applicability threshold are

$$s_{a,i}(x) = q_a(x)^\top k_{a,i}, \quad (10)$$

$$\tau_a(x) = -1 + 2\sigma(xu_a + \beta_a). \quad (11)$$

The evaluated `linear_angular` rule defines

$$c_{a,i}(x) = \text{clip}\left(\frac{s_{a,i}(x) - \tau_a(x)}{\max(1 - \tau_a(x), 10^{-4})}, 0, 1\right). \quad (12)$$

We call $c_{a,i}(x) \in [0, 1]$ the *continuous-valued applicability* of operator i at state x . Its exact applicability support is

$$\begin{aligned} \mathcal{I}_a(x) &= \{i : c_{a,i}(x) > 0\} \\ &= \{i : s_{a,i}(x) > \tau_a(x)\}. \end{aligned} \quad (13)$$

When $s_{a,i}(x) \leq \tau_a(x)$, an operator contributes exactly zero; when $s_{a,i}(x) > \tau_a(x)$, $0 < c_{a,i}(x) \leq 1$ varies continuously with the applicability score. Thus the same applicability score and applicability threshold specify exact non-participation and continuous-valued applicability.

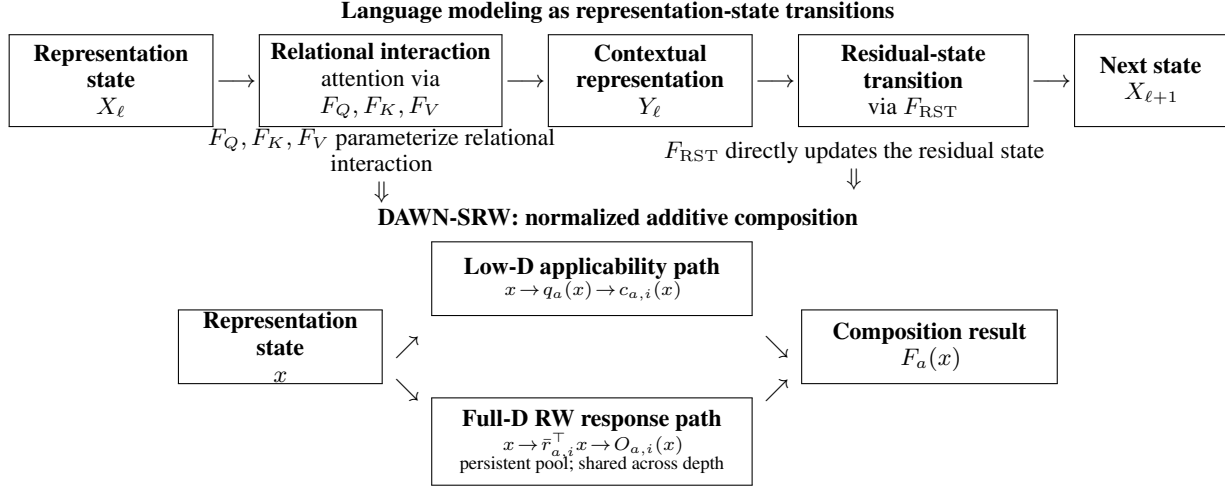


Figure 1. The two-level construction. Top: a language-model block produces state transitions through relational interaction and a subsequent residual-state transition. Bottom: the current state follows two distinct paths. The applicability path uses the low-dimensional operation-address representation to produce $c_{a,i}(x)$; the RW response path reads a scalar $\bar{r}_{a,i}^\top x$ from the full D -dimensional representation and writes $O_{a,i}(x)$ along $\bar{w}_{a,i}$. Their product is a controlled operation contribution, and normalized additive composition produces $F_a(x)$. The type-specific operator pool is persistent across depth, candidate scope is full, and physical RW execution is dense.

3.4. Normalized Additive Composition

DAWN-SRW uses normalized additive composition as one concrete realization of the composition rule Γ . The rule combines controlled RW-operator contributions; $F_a(x)$ denotes the resulting output, not the rule itself. With normalization factor $Z_a(x)$,

$$Z_a(x) = \left[\max \left(\sum_i c_{a,i}(x), 1 \right) \right]^{p_a}, \quad (14)$$

$$F_a(x) = \frac{\sqrt{D/L}}{Z_a(x)} \sum_i c_{a,i}(x) \bar{w}_{a,i} \bar{r}_{a,i}^\top x. \quad (15)$$

In each term, $c_{a,i}(x)$ specifies state-conditioned applicability and $\bar{w}_{a,i} \bar{r}_{a,i}^\top x$ is the operation response; their product is a controlled operation contribution. This additive form is a deliberate realization choice: at a fixed state, it preserves an explicit operator-wise decomposition of the transition-producing output, making individual controlled contributions directly measurable and available as intervention targets; the reported interventions suppress or restore selected numerator terms while leaving the normalization factor unchanged. A nonlinear mixer would generally introduce cross-operation interactions that obscure this exact decomposition; because both $c_{a,i}(x)$ and $Z_a(x)$ are state-dependent, the overall mapping $x \mapsto F_a(x)$ remains generally nonlinear. Exact composition parameters are listed in Appendix A.

3.5. Operation-Address Space

In this realization, evaluating the condition under which an operation contributes becomes a learned geometric relation

between the current state and that operator. The operation-address space is an experimentally controllable inductive bias for state–operation applicability, not part of the State-Conditioned Program definition. Representation and RW computation remain in $\mathbb{R}^D$, while the query and operator addresses $q_a(x), k_{a,i} \in \mathbb{R}^{d_c}$ are control variables. The control projection need only preserve distinctions useful for applicability; once an operation is applicable, Equation 7 still reads the original D -dimensional representation state. Two states can therefore have similar applicability profiles yet different full-dimensional RW read responses and transition contributions. This is an architectural consequence of the equations, not an experimentally established equivalence relation. Here, $d_c = 256$ and

$$q_a(x) = \text{unit}(xP_a + b_a) \in \mathbb{R}^{d_c}. \quad (16)$$

A shared probe pair $P_r, P_w \in \mathbb{R}^{D \times d_c}$ maps functional read and write directions into the operator address

$$k_{a,i} = \text{unit}(\text{unit}(\bar{r}_{a,i}^\top P_r) \odot \text{unit}(\bar{w}_{a,i}^\top P_w)) \in \mathbb{R}^{d_c}. \quad (17)$$

Each operator address is a learned function of what its RW operator reads and where it writes, rather than an independently parameterized address. The operator address is not the scalar operand read by the RW operator: it is a low-dimensional coordinate used to evaluate applicability of the full-dimensional read–write transformation.

Because both $q_a(x)$ and $k_{a,i}$ are unit-normalized, they lie on the unit sphere $\mathbb{S}^{d_c-1}$. Writing $\theta_{a,i}(x)$ for the angle between them, $s_{a,i}(x) = q_a(x)^\top k_{a,i} = \cos(\theta_{a,i}(x))$, so $s_{a,i}(x) > \tau_a(x)$ is equivalent to $\theta_{a,i}(x) < \arccos(\tau_a(x))$. Thus the exact applicability support corresponds geometri-

cally to a state-dependent spherical cap, or angular neighborhood, centered at $q_a(x)$ with angular radius $\arccos(\tau_a(x))$. Within this neighborhood, $c_{a,i}(x)$ gives continuous-valued applicability that varies with the cosine applicability score above the threshold and determines to what degree the operator contributes. This is the spherical realization of $(x, O_{a,i}) \mapsto c_{a,i}(x)$; exact implementation details are restated in Appendix A. The construction induces locality in address geometry, but the present measurements do not establish that geometric proximity corresponds to functional or causal similarity.

3.6. Attention-Mediated and Residual Transitions

For $\tilde{X}_\ell = \text{LN}_\ell^{(1)}(X_\ell)$, three normalized additive compositions produce the attention interfaces:

$$Q_\ell = F_Q(\tilde{X}_\ell),$$

$$K_\ell = F_K(\tilde{X}_\ell),$$

$$V_\ell = F_V(\tilde{X}_\ell),$$

$$A_\ell = \text{softmax}\left(\frac{Q_\ell K_\ell^\top}{\sqrt{d_h}} + M_{\text{causal}}\right),$$

$$\Delta X_{\text{attn},\ell} = A_\ell V_\ell W_\ell^O, \quad (18)$$

$$Y_\ell = X_\ell + \Delta X_{\text{attn},\ell}. \quad (19)$$

Q and K use a shared type-specific operator pool, while V uses a separate operator pool. A second normalization $\tilde{Y}_\ell = \text{LN}_\ell^{(2)}(Y_\ell)$ then queries the RST operator pool:

$$X_{\ell+1} = Y_\ell + F_{\text{RST}}(\tilde{Y}_\ell). \quad (20)$$

Thus RST applicability is evaluated from the contextualized post-attention state, and its normalized additive composition produces the direct residual-state update.

3.7. Persistent Operator Identity and Shared Control

Type-specific operator pools and their state-conditioned control parameterizations are shared across all L layers. Evolving normalized states therefore query the same operator pools rather than a separately parameterized profile at each depth. Persistence makes reuse measurable; it does not establish that broad cross-context reuse emerges. The two LayerNorms and attention output projection W_ℓ^O remain layer-specific; Appendix B gives the complete sharing inventory.

3.8. Candidate Scope and Physical Execution

For computation type a , the candidate scope is the full type-specific operator pool. Thus

$$\mathcal{I}_a(x) \subseteq \{1, \dots, N_a\},$$

with all N_a operators considered as candidates before applicability is determined. Applicability support is generally a

Table 1. Headline language-model results. The 1.3B row is descriptive.

Model	Parameters	C4 CE	Six-task mean
DAWN-SRW 400M	393.8M	2.940213	0.462872
Dense 400M	393.9M	2.935894	0.459652
DAWN-SRW 1.3B	1.272B	2.835879	0.485844

strict subset, while physical RW execution is dense. Sharding, chunking, and accounting details appear in Appendix F.

4. Experimental Setup

The primary model is DAWN-SRW 400M (393.8M parameters, approximately 40B packed C4 tokens), compared with a parameter- and training-token-matched dense Transformer control (393.9M parameters). The 1.3B checkpoint, trained on approximately 20B tokens, is descriptive only. Complete architectural and training settings are in Appendix B.

Performance evaluation uses packed C4 validation and six zero-shot tasks: LAMBADA, HellaSwag, PIQA, ARC-Easy, ARC-Challenge, and WinoGrande. The two 400M models use identical evaluation protocols; complete metrics, splits, and software details are in Appendix C.

For causal analysis, IOI discovery ranks operator sites and freezes a circuit before independent validation and held-out test. Suppression removes the frozen sites’ execution numerator, four matched-control families test specificity, and exact restoration reinserts the selected same-example numerator under the unchanged normalization factor. Appendix D gives the selection criteria, unit counts, statistics, and complete intervention protocol.

5. Results

5.1. Language-model performance

Table 1 consolidates the headline results. At 400M scale, DAWN-SRW differs from the matched dense control by +0.004318918 C4 cross-entropy and +0.003220495 six-task mean. These small measured gaps establish viability through performance retention, not superiority or formal equivalence.

The descriptive 1.3B checkpoint reaches 2.835879 C4 cross-entropy and a 0.485844 six-task mean; without a matched dense control, this is not a scaling comparison. Appendix C reports C4 accuracy and perplexity, the complete task-level breakdown, and standard errors.

5.2. State-Conditioned Exact Applicability Support

At all 9,994,240 packed-C4 positions, Q, K, V, and RST each have strict-subset applicability support according to

Table 2. Exact applicability support according to $c_{a,i}(x) > 0$. Q/K union counts an operator in the applicability support for Q or K at the same layer and token.

Computation	Pool	Mean support	Pool fraction
Q	4,600	487.896	10.6064%
K	4,600	386.722	8.4070%
Q/K union	4,600	812.388	17.6606%
V	14,500	1,649.994	11.3793%
RST	29,300	2,017.703	6.8864%

$c_{a,i}(x) > 0$ (Table 2). This provides evidence of conditional organization through state-conditioned exact applicability support under Equation 12, with strict-subset rather than full-pool participation.

Restricting RW execution to exact applicability support would substantially reduce static accounting, but the resulting estimate remains above the matched dense control; this is not measured latency, energy, or hardware speedup. Detailed accounting and Figure 2 are in Appendix F.

5.3. Frozen IOI circuit and causal intervention

Discovery and confirmation are separated: a 4,096-site Q/K-centered circuit is frozen before validation, and held-out test reuses the unchanged specification. Complete discovery statistics are in Appendix D.

Suppression causes large behavior changes on both splits and exceeds every replicate in all four matched-control families. Activation-matched controls are strongest; frozen-minus-control margin separation is 2.464297 on validation and 1.937754 on test (both $p = 0.00049975$). Exact restoration recovers more than 99.8% of the suppression effect. Together these interventions establish causal accessibility for the evaluated IOI behavior; they do not imply a unique or human-readable algorithm.

5.4. Fail-Closed Results

Fail-closed results on RAVEL, ARC, and 1.3B IOI arise from distinct preregistered criteria: localization instability, failed causal suppression, and insufficient cumulative importance, respectively. Exact statistics and decisions are in Appendix E and Table 9.

6. Discussion

6.1. What the results establish

The constructive test supports three ordered claims. First, the matched 400M comparison establishes viability under the measured protocols. Second, state-conditioned exact applicability support provides evidence of conditional organization in the evaluated model; candidate scope is the

full type-specific operator pool and physical RW execution is dense. Third, the discovery-frozen IOI circuit establishes causal accessibility: suppression changes behavior, matched controls do not reproduce the effect, and exact same-example restoration recovers it on independent validation and held-out data. Fail-closed results on RAVEL, ARC, and 1.3B IOI delimit where this localization procedure did not succeed.

These findings support one explicit realization under the evaluated conditions, not an intrinsic or unique decomposition. Architectural exposure makes operator identities, applicability coefficients, RW responses, and controlled contributions direct measurement and intervention targets. The specified suppression/restoration result establishes causal accessibility for the selected IOI contributions; it does not establish human-readable operator semantics, global modularity, a complete or unique causal decomposition, or transferable program fragments.

Conceptually, the State-Conditioned Program is one program-oriented formulation of the broader problem of explicit learned state-transition organization; its continuous-composition mathematics has strong precedent in dynamical systems, without implying formal operational semantics or mathematical equivalence.

6.2. From the Computational Question to an Open Design Space

Making learned state-transition organization explicit exposes four design axes: control and applicability, operation and identity, coupling and composition, and candidate scope and physical execution. The State-Conditioned Program supplies the current program-oriented formulation, and DAWN-SRW supplies one tested point—rank-1 RW operators, an operation-address space, cross-depth persistence, full candidate scope, dense execution, and normalized additive composition—rather than requirements for the broader design space.

Control and applicability. How should state–operation applicability be represented? DAWN-SRW derives $C_a(x)$ in a learned $d_c = 256$ operation-address space. Alternatives could use functional parameters, intrinsic read activation, separate control representations, or no geometry. The evaluated spherical realization induces state-dependent angular neighborhoods, but it is unestablished whether this geometry predicts functional or causal computation. A central open question is whether learned control geometry recovers a geometry of transition requirements rather than merely a convenient routing substrate.

Operation and identity. What should constitute a reusable learned operation? Rank-1 RW operators are one

Table 3. Frozen IOI circuit results. Accuracy arrows show intact to suppressed. Recovery is the fraction of the suppression effect recovered by exact same-example numerator restoration.

Split	Units	Intact margin	Suppressed margin	Drop (95% CI)	Accuracy	Recovery / restored acc.
Validation	123	3.465456	-4.399331	7.864787 [7.490080, 8.241263]	1.000000 → 0.008130	0.999661 / 0.991870
Held-out test	114	3.254720	-4.155613	7.410332 [7.016485, 7.818472]	1.000000 → 0.008772	0.998406 / 1.000000

granularity; alternatives could be finer, coarser, structured, or hierarchical. DAWN-SRW persists type-specific identities across depth, while other scopes could be layer-local or recurrent. Persistence makes reuse measurable; it does not establish that broad reuse emerges, guarantee a single function, or imply semantic purity. Operations may remain distributed or polysemantic.

Coupling and composition. After control and operations are made explicit, how should their controlled contributions combine into a transition? DAWN-SRW derives addresses from read/write parameters without identifying control variables with RW operations, multiplies each response by its applicability, and uses normalized additive composition partly to preserve exact operator-wise terms. Interactive or nonlinear composition could trade separability for cross-operation structure; sequential, hierarchical, recurrent, or more tightly coupled designs remain possible. Whether an exactly additive decomposition is necessary for useful causal access is itself an open question.

Candidate scope and physical execution. How much of an operator pool should candidate scope include? DAWN-SRW uses the full type-specific operator pool. Restricted or state-dependent retrieval, sparse candidate selection, and sparse physical RW execution pose distinct compute/quality questions; none is established by exact applicability support.

6.3. Structural Questions Exposed by the Organization

These questions concern explicit learned transition organization generally; the present DAWN-SRW geometry and language-model evidence instantiate but do not settle them.

Dimension. The architecture distinguishes the representation dimension D , control dimension d_c , the local span of an RW update, and the full operator pool. The evaluated model is viable with $d_c = 256 \ll D$, while RW responses still read and write in full D . This does not establish that d_c is minimal or sufficient, that useful local execution is generally low-dimensional, or that computation-selection complexity is intrinsically separate from state-content complexity. A concrete question is how much of the ambient representation space the applicable write directions and the full learned population must span.

Regime and geometry. Do applicability profiles define overlapping computational regimes, and does proximity in control geometry predict similar transition requirements or causal effects? The present study measures exact applicability support but not regime equivalence, program similarity, or geometry–function correspondence.

Reuse and composition. Persistent identities could become state-specific specialists, locally reused operations, cross-context reusable operations, or recurring higher-order motifs. Which organization emerges is unmeasured. It is likewise open whether applicability depends primarily on what an operation reads, on the effect it writes, or on their joint read–write identity; the present architecture uses the joint construction but does not compare alternatives.

Causal invariance and counterfactual intervention. The present causal protocol validates suppression and exact same-example restoration of one discovery-frozen circuit. Whether similarity in state-conditioned operation organization predicts similar intervention effects, and whether control-profile transplant, operation transfer, or counterfactual transition execution is meaningful, remain future questions rather than demonstrated capabilities.

6.4. From Applicability Support to Physical Execution

For a given representation state, many operators have exactly zero applicability, but this does not establish restricted candidate scope or sparse physical execution. In DAWN-SRW,

$$\mathcal{I}_a(x) \subseteq \{1, \dots, N_a\},$$

so the candidate scope is the full type-specific operator pool and physical RW execution is dense. Applicability support specifies which operators contribute; candidate scope specifies which operators control considers; physical execution specifies how computation maps to hardware. **Sparse applicability support does not by itself imply hardware-exploitable sparse computation.** The open problem is to restrict candidate scope and physical execution while preserving language-model quality and conditional organization.

6.5. Limitations

Global candidate evaluation and dense RW execution remain expensive, and performance may depend partly on the

capacity of the full operator pool. Even static accounting with RW execution restricted to exact applicability support remains 2.1168 times the matched dense control’s backbone. No production latency, energy, or hardware advantage is measured.

The familiar state–condition–operation motif is not claimed as novel, and an arbitrary post-hoc functional decomposition does not establish computational organization. The study does not establish that the State-Conditioned Program organization or its control–operation factorization is intrinsic, unique, necessary, privileged, or optimal. Nor are the spherical geometry, $d_c = 256$ control dimension, or correspondence between geometric and functional organization shown to be intrinsic. In particular, the low-dimensional applicability path is not shown to be minimal, sufficient, a statistical quotient of the representation, or evidence that local execution is generally low-dimensional; the geometry may instead be a high-capacity control substrate for broadly distributed full-dimensional operations. Persistence of operator identity likewise makes reuse measurable but does not establish broad cross-context reuse.

Causal evidence is confined to the specified IOI setting. Fail-closed results on RAVEL, ARC, and 1.3B IOI delimit mechanistic generalization; the differently budgeted 400M and 1.3B checkpoints do not define a controlled scaling curve; and individual operators need not map one-to-one to human-readable concepts. Suppression/restoration establishes causal accessibility only for the specified terms and behavior; it does not establish arbitrary program editing, semantic interpretability, operator replacement, or control transplant.

7. Conclusion

Learned state-transition computation is usually organizationally implicit. The coordinates $\mathcal{P} = (\mathcal{O}, C, \Gamma)$ provide a minimal description of one family of explicit organizations, not a universal or necessary decomposition. A State-Conditioned Program is one program-oriented formulation requiring these roles to participate in actual forward execution. The architectural hypothesis is that learned transition-producing computation can realize such organization through stable learned operations under explicit state-conditioned control over their contributions. DAWN-SRW tests one concrete rank-1 RW realization in the present constructive domain of autoregressive language modeling: applicability is computed in $d_c = 256$, while each operation reads a scalar from and writes a direction into the full D -dimensional representation.

The evidence is bounded: measured 400M viability, descriptive 1.3B results, evidence of conditional organization from state-conditioned exact applicability support, and causal accessibility through a discovery-frozen IOI circuit tested by suppression, matched controls, and exact restoration. Fail-closed results mark where localization did not succeed.

The main interpretation is that one learned high-dimensional state-transition system can expose referable operation contributions while retaining viability, not that its decomposition or geometry is intrinsic, unique, semantically pure, or computationally optimal. The broader question is how learned state-transition computation can retain the flexibility of continuous neural representations while organizing the conditions, operations, and transition effects that constitute its execution as explicit objects of measurement and intervention.

A. Exact Operation-Address and Composition Equations

This appendix restates the production computation as a compact reference. Raw read and write parameters are normalized in float32 before being cast for dense contractions:

$$\bar{r}_i = \frac{r_i}{\max(\|r_i\|_2, \epsilon)}, \quad \bar{w}_i = \frac{w_i}{\max(\|w_i\|_2, \epsilon)}.$$

One pair of model-level probes is shared across QK, V, and RST. In canonical notation, the operator address is

$$k_{a,i} = \text{unit}(\text{unit}(\bar{r}_{a,i}^\top P_r) \odot \text{unit}(\bar{w}_{a,i}^\top P_w)).$$

The query and applicability threshold use separate direct projections. The `DirectTau` projection produces the implementation variable $\text{raw_tau} = xu_a + \beta_a$, whose transformed value is the canonical applicability threshold $\tau_a(x)$:

$$q_a(x) = \text{unit}(xP_a + b_a), \\ \tau_a(x) = -1 + 2\sigma(xu_a + \beta_a), \quad s_{a,i}(x) = q_a(x)^\top k_{a,i}.$$

For the `linear\_angular` applicability rule:

$$c_{a,i}(x) = \text{clip}\left(\frac{s_{a,i}(x) - \tau_a(x)}{\max(1 - \tau_a(x), 10^{-4})}, 0, 1\right)$$

and padding rows are masked to zero. This is the continuous-valued applicability used in normalized additive composition. With normalization factor $Z_a(x) = \max(\sum_i c_{a,i}(x), 1)^{p_a}$, the production composed output before causal attention or residual addition is

$$F_a(x) = \sqrt{\frac{D}{L}} \frac{\sum_i c_{a,i}(x) \langle x, \bar{r}_{a,i} \rangle \bar{w}_{a,i}}{Z_a(x)},$$

with $p_Q = p_K = 0.5$, $p_V = 1.0$, and $p_{\text{RST}} = 1.2$. Operator rows are sharded over the model axis; local numerators and local sums of applicability values are reduced across shards to produce the global numerator and normalization factor. Chunking changes the contraction schedule, not the equation.

Local Jacobian and update-span consequence. Fix x and abbreviate $\alpha = \sqrt{D/L}$, $c_i = c_{a,i}(x)$, $u_i = \bar{r}_{a,i}^\top x$, $Z = Z_a(x)$, and $g_i = c_i u_i / Z$. On a neighborhood with fixed exact applicability support, and at points away from clipping, support, and normalization kinks, the realized normalized RW composition has

$$\nabla_x g_i = Z^{-1}(c_i \bar{r}_{a,i} + u_i \nabla_x c_i) - Z^{-2} c_i u_i \nabla_x Z, \\ J_{F_a}(x) = \alpha \sum_{i \in \mathcal{I}_a(x)} \bar{w}_{a,i} \nabla_x g_i^\top.$$

For $\mathcal{W}_a(x) = \text{span}(\{\bar{w}_{a,i}\}_{i \in \mathcal{I}_a(x)})$, consequently

$$\text{Im } J_{F_a}(x) \subseteq \mathcal{W}_a(x), \quad \text{rank } J_{F_a}(x) \leq \dim \mathcal{W}_a(x).$$

For a residual transition $T(x) = x + F_a(x)$, the same containment applies to $J_T(x) - I$, not generally to $J_T(x)$. This is a local consequence of the evaluated normalized additive RW realization under the stated regularity conditions, not a theorem about every State-Conditioned Program.

B. Complete Model and Training Settings

The 400M DAWN-SRW checkpoint is the primary performance and mechanistic model. The dense 400M checkpoint is only a matched performance control. The 1.3B DAWN-SRW checkpoint is used for descriptive performance and one IOI localization replication that produced a fail-closed result. The still-incomplete dense 1.3B run is excluded.

The QK, V, and RST operator pools; the two RW-to-address probes; the Q/K/V query projection; the RST query projection; and their `DirectTau` projections are shared across all layers. Q and K share read/write rows but retain separate query slices, applicability thresholds, applicability values, and outputs. The two LayerNorms and attention output projection W_ℓ^O are layer-specific; positional and token embeddings and the final normalization are model-level parameters. Depth is not an explicit operator-pool index: it changes the representation state presented to the shared control interface.

C. Benchmark Metrics and Protocol Details

C4. All reported C4 rows use the same packed validation artifact and `packed_c4_validation_v1` protocol. The capped view contains 19,531 sequences. Exactly 19,520 sequences form 610 complete global evaluation batches, producing 9,974,720 valid next-token targets; 11 incomplete-tail sequences are excluded before scoring.

Zero-shot. The protocol is `mamba_table3_zero_shot_v1`, using stock `lm-eval==0.4.2`, global batch 32, maximum length 512, length buckets 64/128/256/512, and zero few-shot examples. Evaluated split sizes are 5,153 for LAMBADA, 10,042 for HellaSwag, 1,838 for PIQA, 2,376 for ARC-Easy, 1,172 for ARC-Challenge, and 1,267 for WinoGrande. Task, dataset, tokenizer, and software settings are identical between the two 400M models.

C.1. Detailed benchmark results

LAMBADA perplexities are 24.0774 ± 0.8005 for DAWN-SRW 400M, 24.0042 ± 0.8073 for dense 400M, and 17.6358 ± 0.5390 for DAWN-SRW 1.3B.

Table 4. Model and training specification. The 1.3B row is descriptive and does not form a controlled scaling curve with the 400M rows.

Model	Parameters	Tokens	D	Layers	Heads	QK / V / RST rows	Checkpoint
DAWN-SRW 400M	393,800,708	39,999,504,384	2,304	18	36	4,600 / 14,500 / 29,300	76,293
Dense Transformer 400M	393,896,960	39,999,504,384	1,280	18	20	—	76,293
DAWN-SRW 1.3B	1,272,038,404	19,999,981,568	4,096	24	32	8,534 / 26,904 / 54,352	87,193

Table 5. Detailed settings from the registered training configurations and restored checkpoint manifests.

Setting	DAWN-SRW 400M	Dense 400M	DAWN-SRW 1.3B
Tokenizer vocabulary	30,522	30,522	30,522
Residual width	2,304	1,280	4,096
Layers	18	18	24
Heads / head width	36 / 64	20 / 64	32 / 128
Maximum sequence length	512	512	512
Feed-forward width	—	5,120	—
Control dimension d_c	256	—	256
QK / V / RST operator-pool size	4,600 / 14,500 / 29,300	—	8,534 / 26,904 / 54,352
QK / V / RST chunks	1 / 1 / 4	—	1 / 2 / 8
QK / V / RST normalizer exponent p_a	0.5 / 1.0 / 1.2	—	0.5 / 1.0 / 1.2
Initial applicability-support fraction targets	0.05 / 0.03 / 0.01	—	0.05 / 0.03 / 0.01
Training batch	1,024	1,024	448
Mesh (data $\times$ model)	16×2	16×2	16×2
Learning rate	3×10^{-4}	3×10^{-4}	2×10^{-4}
Warmup ratio	0.08	0.08	0.06
Weight decay	0.1	0.1	0.1
Applicability-threshold learning-rate multiplier	0.001	—	0.001
Dropout / <code>router.dropout</code>	0 / 0	0 / —	0 / 0
Gradient checkpointing	yes	yes	yes
Seed	1	1	1

Table 6. Packed C4 validation on the identical 9,974,720 target tokens.

Model	CE	PPL	Accuracy
DAWN-SRW 400M	2.940213	18.919873	0.424912
Dense 400M	2.935894	18.838336	0.427100
DAWN-SRW 1.3B	2.835879	17.045382	0.437099

D. IOI Localization and Intervention Details

D.1. Frozen discovery

Discovery admits 119 paired-correct independent units and ranks 456,697 sites. Split-half rank stability is 0.913978765. The frozen selection rule is the smallest prefix satisfying cumulative absolute importance at least 0.70 and split top- k overlap at least 0.95. The selected $k = 4096$ prefix has:

- cumulative absolute importance 0.719248713;
- 3,993 shared split sites, overlap 0.974853516, and Jac-card 0.950940700;
- Q/K/V/RST counts = 1744/2350/2/0;
- coverage of every layer from 0 through 17.

Validation and test do not recompute or alter this selection.

D.2. Suppression, controls, and restoration

Let T_{corr} and T_{src} denote the correct-answer and source-answer token sets. The primary score is

$$\mathcal{M} = \sum_{t \in T_{\text{corr}}} \log p(t) - \sum_{t \in T_{\text{src}}} \log p(t).$$

Suppression zeros the frozen sites’ execution numerator at specified token positions. It leaves applicability and the normalization factor unchanged. Controls use the same 4,096-site count and exclude all circuit sites. Activation matching conditions on layer, computation type, and the discovery mean pre-scale operator-output-norm quantile. Restoration executes a distinct intervention: after suppression, it adds back the exact selected same-example numerator under the unchanged normalization factor.

On validation, restoration returns the margin to 3.462791, leaving residual drop 0.002665. On held-out test, it returns the margin to 3.242905, leaving residual drop 0.011815. The respective recovery fractions are 0.999661155 (95% CI [0.998654544, 1.000584125]) and 0.998405645 (95% CI [0.997323737, 0.999385266]).

E. Fail-Closed Results

RAVEL. Discovery uses 142 independent units. Rank stability is 0.628762 for Continent, 0.676243 for Country, and 0.660622 for Language, below the preregistered requirement. No prefix passes the complete discovery rule, no variable circuit is frozen, and validation, transfer, and test are not opened.

ARC. Discovery stability is 0.89386 and a 4,096-site Q/K-only circuit is frozen. On 51 validation units, intact margin

Table 7. Zero-shot accuracy. Parentheses contain standard errors. HS, AE, and AC denote HellaSwag, ARC-Easy, and ARC-Challenge.

Model	LAMBADA	HS	PIQA	AE	AC	WinoGrande	Mean
DAWN-SRW 400M	0.352028 (0.006654)	0.458375 (0.004972)	0.712187 (0.010563)	0.484007 (0.010255)	0.256826 (0.012767)	0.513812 (0.014047)	0.462872
Dense 400M	0.353192 (0.006659)	0.457080 (0.004971)	0.698041 (0.010712)	0.475589 (0.010248)	0.249147 (0.012639)	0.524862 (0.014035)	0.459652
DAWN-SRW 1.3B	0.395886 (0.006813)	0.510157 (0.004989)	0.723069 (0.010440)	0.509680 (0.010258)	0.268771 (0.012955)	0.507498 (0.014051)	0.485844

Table 8. Control effects and frozen-minus-control separation. Each control family contains 100 replicates. All paired permutation p -values are 0.0004997501 with 2,000 permutations.

Control	Validation drop	Frozen minus control (95% CI)	Test drop	Frozen minus control (95% CI)
Equal-count	0.070886	7.793902 [7.437851, 8.161572]	0.080276	7.330057 [6.922408, 7.716879]
Same-layer	0.039389	7.825399 [7.440629, 8.180288]	0.041073	7.369260 [6.977447, 7.754539]
Activation-matched	5.400490	2.464297 [2.076200, 2.832697]	5.472579	1.937754 [1.564818, 2.319706]
Route-frequency	0.219162	7.645626 [7.289512, 8.031625]	0.242665	7.167668 [6.779836, 7.545537]

Table 9. Fail-closed results. Later phases were not opened after a failed preregistered criterion.

Study	Passed	Failed criterion	Decision
RAVEL 400M	Behavioral eligibility	Variable stability: 0.628762 / 0.676243 / 0.660622	No frozen circuit; no validation, transfer, or test
ARC 400M	Stable discovery; 4,096 Q/K sites; restoration	Suppression effect wrong sign; control separation failed	Reject causal replication; held-out test closed
IOI 1.3B	Behavioral eligibility; rank stability 0.907188	4,096-site cumulative importance 0.635144 < 0.70	No frozen circuit; validation and test closed

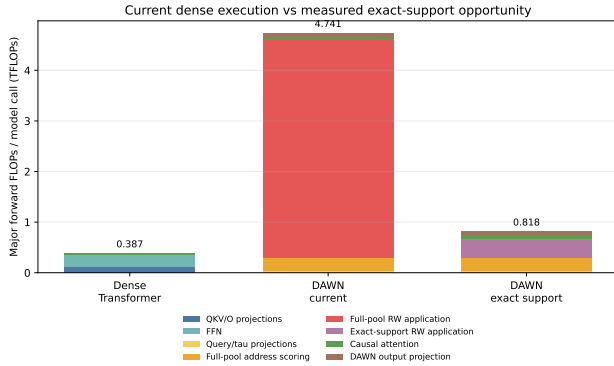


Figure 2. Static backbone accounting at batch 1 and sequence length 512. Address scoring remains dense in the estimate restricted to exact applicability support. Current dense RW execution also applies all full-dimensional RW rows. No production latency claim is made.

is 1.735534 and suppressed margin is 2.819139, giving a frozen margin drop of -1.083605 (95% CI $[-1.472180, -0.690348]$). The effect has the wrong sign and fails control separation. Restoration recovery is 1.004083 (95% CI $[0.994587, 1.013652]$), but restoration alone does not pass the conjunction. Held-out test remains closed.

1.3B IOI. The scale replication admits 125 independent units and passes rank stability at 0.907187749. The largest allowed 4,096-site prefix reaches only 0.635144 cumulative absolute importance, below the 0.70 criterion. No circuit is frozen; validation and test remain closed.

F. Compute Accounting

The implementation shards operator rows across the model mesh and processes them in dense chunks. The 400M configuration uses one QK chunk, one V chunk, and four RST chunks. Every query scores all local operator addresses, and every operator row participates in dense read and write contractions before cross-shard reduction. RW operators execute in the full residual dimension D ; the d_c -dimensional operation-address space is used only for applicability.

The applicability-support analysis evaluates 9,994,240 packed-C4 positions and observes applicability scores and applicability thresholds before RW application. It stores aggregate statistics, not per-token operator identities or vectors. Static accounting uses batch 1, sequence length 512, and 1 MAC = 2 FLOPs. It counts major matmul/einsum operations and omits softmax, normalization, applicability elementwise arithmetic, embedding lookup, residual additions, and communication collectives.

Static forward accounting is 4.741260116 TFLOPs for current dense DAWN-SRW, 0.818230250 TFLOPs for dense addressing plus RW application restricted to exact applicability support, and 0.386547057 TFLOPs for the matched dense control. The estimate restricted to exact applicability support is 17.2577% of current DAWN-SRW accounting but still 2.1168 times the dense Transformer. These are static arithmetic estimates, not measured latency, energy, or hardware speedup.

The estimate restricted to exact applicability support is mathematically output-equivalent if the executor can skip all and

Table 10. Backbone static accounting. “Exact applicability support” retains full dense address scoring and replaces only full-pool RW application with the measured applicability support according to $c_{a,i}(x) > 0$.

Component	Current dense DAWN-SRW	Dense address + exact RW	Dense Transformer
Address scoring + query/threshold projections	0.293742	0.293742	–
RW application	4.306187	0.383157	–
Causal attention	0.043487	0.043487	0.024159
Dense output / QKV / FFN maps	0.097845	0.097845	0.362388
Total TFLOPs	4.741260	0.818230	0.386547

only zero-weight RW terms. It does not include indexed address retrieval and does not demonstrate kernel parity. Instrumented profiling time is excluded from latency claims because histogram collection is part of the measured path.

G. Reproducibility Identities

The experiments reported here use DAWN-SRW implementation revision v4.1.7.2, recorded under the repository model identity `spatial-rl-v4.1.7.2`; literal repository filenames and artifact paths retain the historical shorthand v4172.

The machine-readable checkpoint registry is `configs/paper_checkpoint_registry.yaml`. Exact evaluation commits, software versions, data hashes, result roots, and complete artifact hashes are recorded in the repository experiment ledger. Historical experiment branches are evidence identities only; this source is maintained on the canonical main branch.

Acknowledgments

The author thanks Anthropic Claude, Claude Code, OpenAI ChatGPT, and OpenAI Codex for assistance throughout this research. This work was developed with extensive use of AI systems for literature exploration, conceptual exploration, cross-domain framing, abstraction and terminology refinement, implementation support, debugging, and manuscript preparation. In particular, these systems were used as interactive assistants for refining the learned state-transition formulation and the exposition of DAWN-SRW. The research direction, architectural decisions, experimental design, experiments, analyses, interpretations, and final claims remain the responsibility of the author.

This research was supported by the Google TPU Research Cloud (TRC) program, which provided computational resources for training and analysis.

References

Berges, V.-P., Oğuz, B., Haziza, D., Yih, W.-t., Zettlemoyer, L., and Ghosh, G. Memory layers at scale. *arXiv preprint arXiv:2412.09764*, 2024. doi: 10.48550/arXiv.2412.09764.

Cases, I., Rosenbaum, C., Riemer, M., Geiger, A., Klinger, T., Tamkin, A., Li, O., Agarwal, S., Greene, J. D., Jurafsky, D.,

Potts, C., and Karttunen, L. Recursive routing networks: Learning to compose modules for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pp. 3631–3648. Association for Computational Linguistics, 2019. doi: 10.18653/v1/N19-1365.

Chen, Y., Gu, N., Shang, J., Zhang, Z., Feng, Y., Sheng, J., Liu, T., Wang, S., Sun, Y., Wu, H., and Wang, H. Mixture of universal experts: Scaling virtual width via depth-width transformation. *arXiv preprint arXiv:2603.04971*, 2026. doi: 10.48550/arXiv.2603.04971.

Csordás, R., Irie, K., Schmidhuber, J., Potts, C., and Manning, C. D. MoEUT: Mixture-of-experts universal transformers. In *Advances in Neural Information Processing Systems*, volume 37, 2024a. doi: 10.52202/079017-0897.

Csordás, R., Piekos, P., Irie, K., and Schmidhuber, J. SwitchHead: Accelerating transformers with mixture-of-experts attention. In *Advances in Neural Information Processing Systems*, volume 37, 2024b. doi: 10.52202/079017-2368.

Dehghani, M., Gouws, S., Vinyals, O., Uszkoreit, J., and Kaiser, L. Universal transformers. In *International Conference on Learning Representations*, 2019.

Didolkar, A., Goyal, A., Ke, N. R., Blundell, C., Beaudoin, P., Heess, N., Mozer, M., and Bengio, Y. Neural production systems. In *Advances in Neural Information Processing Systems*, volume 34, pp. 25673–25687, 2021.

Fedus, W., Zoph, B., and Shazeer, N. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022.

Geva, M., Schuster, R., Berant, J., and Levy, O. Transformer feed-forward layers are key-value memories. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pp. 5484–5495, 2021. doi: 10.18653/v1/2021.emnlp-main.446.

Geva, M., Caciularu, A., Wang, K., and Goldberg, Y. Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 30–45, 2022. doi: 10.18653/v1/2022.emnlp-main.3.

Goyal, A., Lamb, A., Gampa, P., Beaudoin, P., Blundell, C., Levine, S., Bengio, Y., and Mozer, M. Factorizing declarative and procedural knowledge in structured, dynamical environments. In *International Conference on Learning Representations*, 2021a. URL <https://openreview.net/forum?id=VVdmjgu7pKM>.

Goyal, A., Lamb, A., Hoffmann, J., Sodhani, S., Levine, S., Bengio, Y., and Schölkopf, B. Recurrent independent mechanisms. In *International Conference on Learning Representations*, 2021b. URL <https://openreview.net/forum?id=mLcmdlEUxy->.

He, X. O. Mixture of a million experts. *arXiv preprint arXiv:2407.04153*, 2024. doi: 10.48550/arXiv.2407.04153.

Huang, J., Wu, Z., Potts, C., Geva, M., and Geiger, A. RAVEL:

Table 11. Primary immutable identities. Branch names are omitted because they move; exact checkpoint steps, hashes, and artifact locations are the binding identities.

Evidence	Exact identity	Repository source
DAWN-SRW 400M	model spatial-r1-v4.1.7.2; step 76,293; checkpoint identity a7ce8afc...f91c17; config hash 08733ae4...5039b	configs/train_config_v4172_400M_c4_40B_v4_64_ver1_den_qk0p5_vlp0_rst1p2.yaml
Dense 400M	model baseline-JAX; step 76,293; config hash c3ce5d72...82fc25	configs/train_config_baseline_tpu_400M_c4_40B_v4_64_tp2.yaml
DAWN-SRW 1.3B	model spatial-r1-v4.1.7.2; step 87,193; checkpoint identity 2e38b41f...bf5f9; config hash 1e877263...4a5222	configs/train_config_v4172_1p3B_c4_20B_v4_64_ver1_den_qk0p5_vlp0_rst1p2.yaml
IOI circuit	parsed frozen-spec hash 6ec26618...418c; site hash 573cfff8...0200; selected-row hash 1439f6f3...fe0c1	configs/paper_ioi_frozen_circuit_v4172_400m.yaml
Compute support	support hash b1eac5f5...bdc2; accounting hash d7ad6115...984e	assets/paper/v4172_400m_compute_support_step76293/

- Evaluating interpretability methods on disentangling language model representations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 8669–8687, 2024. doi: 10.18653/v1/2024.acl-long.470.
- Huang, M., Shi, H., Zheng, C., Wu, Y., Chen, G., Yu, X., Yin, Y., and Cheng, H. UniPool: A globally shared expert pool for mixture-of-experts. *arXiv preprint arXiv:2605.06665*, 2026. doi: 10.48550/arXiv.2605.06665.
- Johansen, T. A. and Foss, B. A. Identification of non-linear system structure and parameters using regime decomposition. *Automatica*, 31(2):321–326, 1995. doi: 10.1016/0005-1098(94)00096-2.
- Kramár, J., Lieberum, T., Shah, R., and Nanda, N. AtP*: An efficient and scalable method for localizing LLM behaviour to components. *arXiv preprint arXiv:2403.00745*, 2024. doi: 10.48550/arXiv.2403.00745.
- Lample, G., Sablayrolles, A., Ranzato, M., Denoyer, L., and Jégou, H. Large memory layers with product keys. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Lu, H., Zhao, C., Xue, M., Yao, L., Moore, K., and Gong, D. Little by little: Continual learning via incremental mixture of rank-1 associative memory experts. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, 2026. arXiv:2506.21035.
- Mittal, S., Raparthy, S. C., Rish, I., Bengio, Y., and Lajoie, G. Compositional attention: Disentangling search and retrieval. In *International Conference on Learning Representations*, 2022.
- Mudrik, N., Chen, Y., Yezerets, E., Rozell, C. J., and Charles, A. S. Decomposed linear dynamical systems (dLDS) for learning the latent components of neural dynamics. *Journal of Machine Learning Research*, 25(59):1–44, 2024.
- Neelakantan, A., Le, Q. V., and Sutskever, I. Neural programmer: Inducing latent programs with gradient descent. In *International Conference on Learning Representations*, 2016.
- Nguyen, T., Mansouri, A., Madan, K., Nguyen, K. D., Ahuja, K., Liu, D., and Bengio, Y. Reusable slotwise mechanisms. In *Advances in Neural Information Processing Systems*, volume 36, pp. 23533–23556, 2023. doi: 10.52202/075280-1021.
- Plotkin, G. D. A structural approach to operational semantics. Technical Report DAIMI FN-19, Computer Science Department, Aarhus University, Aarhus, Denmark, September 1981.
- Poli, M. et al. Neural hybrid automata: Learning dynamics with multiple modes and stochastic transitions. In *Advances in Neural Information Processing Systems*, volume 34, pp. 9977–9989, 2021.
- Rahaman, N., Gondal, M. W., Joshi, S., Gehler, P. V., Bengio, Y., Locatello, F., and Schölkopf, B. Dynamic inference with neural interpreters. In *Advances in Neural Information Processing Systems*, volume 34, pp. 10985–10998, 2021.
- Reed, S. and de Freitas, N. Neural programmer-interpreters. In *International Conference on Learning Representations*, 2016.
- Rosenbaum, C., Klinger, T., and Riemer, M. Routing networks: Adaptive selection of non-linear functions for multi-task learning. In *International Conference on Learning Representations*, 2018. URL <https://openreview.net/forum?id=ry8dvM-R->.
- Rugh, W. J. and Shamma, J. S. Research on gain scheduling. *Automatica*, 36(10):1401–1425, 2000. doi: 10.1016/S0005-1098(00)00058-3.
- Schlag, I., Irie, K., and Schmidhuber, J. Linear transformers are secretly fast weight programmers. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 9355–9366, 2021.
- Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., and Dean, J. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *ICLR*, 2017.
- Shen, Y., Zhang, Z., Cao, T., Tan, S., Chen, Z., and Gan, C. ModuleFormer: Modularity emerges from mixture-of-experts. *arXiv preprint arXiv:2306.04640*, 2023. doi: 10.48550/arXiv.2306.04640.
- Shi, J., Peng, Z., Zhu, Y., Wu, Y., Liu, G., and Luo, Y. OmniMoE: An efficient MoE by orchestrating atomic experts at scale. *arXiv preprint arXiv:2602.05711*, 2026. doi: 10.48550/arXiv.2602.05711.
- Takagi, T. and Sugeno, M. Fuzzy identification of systems and its applications to modeling and control. *IEEE Transactions on Systems, Man, and Cybernetics*, SMC-15(1):116–132, 1985. doi: 10.1109/TSMC.1985.6313399.
- Wang, K., Variengien, A., Conmy, A., Shlegeris, B., and Steinhardt, J. Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations*, 2023.
- Weiss, G., Goldberg, Y., and Yahav, E. Thinking like transformers. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 11080–11090, 2021.
- Weiss, M., Rahaman, N., Locatello, F., Pal, C., Bengio, Y., Schölkopf, B., Li, E. L., and Ballas, N. Neural attentive circuits. In *Advances in Neural Information Processing Systems*, volume 35, pp. 7741–7754, 2022. doi: 10.52202/068431-0562.
- Yang, B., Bender, G., Le, Q. V., and Ngiam, J. CondConv: Conditionally parameterized convolutions for efficient inference. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Yang, Y., Wang, C., and Li, J. UMoE: Unifying attention and FFN with shared experts. In *Advances in Neural Information Processing Systems*, volume 38, 2025.
- Zadeh, L. A. Fuzzy sets. *Inf. Control*, 8(3):338–353, 1965.